

# Practice Quiz: Action (Implementation)

Name: \_\_ Date: \_\_

## Part A: Multiple Choice

1. Multi-step policies are generated using: A) Random sampling B) Cartesian product of actions across time steps C) Neural networks D) Manual specification only
2. For 4 actions and 3 time steps, the total number of policies is: A) 12 B) 64 ( $4^3$ ) C) 7 D) 4
3. The pragmatic component of EFE measures: A) How much the agent will learn B) How well expected future observations match the agent's preferences (C vector) C) The speed of action execution D) The number of states
4. The epistemic component of EFE measures: A) The agent's knowledge level B) How much mutual information there is between future states and observations under a policy C) The reward value D) The prior probability
5. In the active inference loop, belief transition is done: A) Before observing B) After executing the action — predicting the next state before the next observation C) Only during learning D) Never
6. The log dictionary in the loop stores: A) Debug messages B) Belief, action, observation, and EFE histories for post-hoc analysis C) Error messages only D) File paths
7. If gamma is very high, the agent: A) Acts randomly B) Commits strongly to the policy with lowest EFE C) Never acts D) Ignores observations

## Part B: Short Answer

1. Explain the computational scaling problem of policy evaluation: how does the number of policies grow with `num_actions` and `policy_length`? At what point does this become impractical?

2. Write pseudocode for selecting an action in the active inference loop. Include state inference, policy evaluation, softmax, and sampling.
3. An agent in a 3-state world has two policies. Policy 1 has pragmatic value 5 and epistemic value 1. Policy 2 has pragmatic value 2 and epistemic value 6. With  $\gamma = 1.0$ , compute  $P(\pi_1)$  and  $P(\pi_2)$  using softmax.